

Ts. MOHAMAD ALIF IRFAN

SOFTWARE ENGINEER | FULL STACK DEVELOPER | SYSTEM ANALYST

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SUMMARY

Software Engineer with Certified Professional Technologist (Ts.) registered under the Malaysia Board of Technologists (MBOT) with over 5 years of experience architecting enterprise solution workflows, customizing ERP platforms (Epicor/Kinetic), and engineering full-stack applications. Proven track record in translating complex business requirements into scalable technical systems using Linux, C++, C#, ASP.NET, Python, Java, Microsoft SQL, Oracle, MySQL and SQL Server. Skilled in end-to-end system integration, cloud infrastructure (AWS, Cloudflare), and automated data pipelines with Power BI and Power Automate. Recognized with an Appreciation Award for leading a seamless, enterprise-wide ERP migration.

CORE COMPETENCIES / TECHNICAL SKILLS

Full-Stack & Mobile (Web apps, PWAs & Native Mobile)

- C#, VB.NET, ASP.NET / .NET MAUI, PHP, Python, Progressive Web Apps (PWA), REST API Design, JavaScript, Java, Tailwind CSS, Swift, C++, C, Next.js / HTML5

Analysis & Process Tools (Systems analysis, automation & BI)

- System & Business Analysis, Microsoft Power Automate, Power BI Data Dashboards, Tableau, Jira & Lucidchart, Bizagi & MS Visio

Database Engineering (Relational & query optimization)

- Microsoft SQL Server, T-SQL (Procedures & Triggers), MySQL, SQL Server Profiler, Database Engine Tuning, Clustered/Non-Clustered Indexes, Supabase, Firebase, Oracle RDBMS, AWS RDS Persistence

Cloud, IoT & DevOps (Edge telemetry & infrastructure)

- Linux / Ubuntu, Raspberry Pi Edge IoT, Modbus / RS485 Sensors, AWS (EC2, S3, RDS), Cloudflare (DNS & CDN), IIS Web Server, CI/CD (Jenkins, Bitbucket), UNIX-Solaris

Enterprise ERP (Kinetic Epicor)

- BAQs, External BAQs, Report Data Definition, UD Tables, UD Fields, BPM Workflows, Epicor Functions, SSRS Report, MES

PROFESSIONAL WORK EXPERIENCE

Software Engineer **May 2024 - Present**

Thong Guan Plastic Technologies Sdn. Bhd. | Sungai Petani, Kedah, Malaysia

- **Enterprise ERP Customization (Epicor / Kinetic):** Tailored core ERP modules, including Purchasing, automated ISCC compliance form generation, and auto-signature embedding for Shipping reports. Engineered custom business logic, automated workflows, and executive reports using BAQs, BPMs, SSRS, and Epicor Customization Frameworks.
- **Full-Stack & Cross-Platform Development:** Engineered end-to-end web, mobile, desktop, and Progressive Web Applications (PWA) across Web, Android, and iOS platforms using C#, .NET MAUI, ASP.NET, JavaScript, Java, and REST APIs, backed by relational and cloud databases including Microsoft SQL Server, MySQL, and AWS cloud database services.

- **IoT & Cloud Integration (IPAK5 Project):** Co-developed the iPak5 integration system from scratch, deploying Raspberry Pi edge devices running Linux to execute Python scripts and REST APIs for real-time production line data acquisition and direct telemetry streaming to AWS cloud infrastructure.
- **Database Engineering & T-SQL Development:** Designed, configured, and optimized relational databases across Microsoft SQL Server, MySQL, and AWS EC2 server environments. Developed complex T-SQL code including stored procedures, indexed views, table functions, and database triggers while optimizing performance via clustered/non-clustered index tuning, execution plan analysis, SQL Server Profiler, and Database Engine Tuning Advisor.
- **Cloud & Hosting Infrastructure:** Configured and managed application deployments on IIS, Cloudflare (DNS, security, CDN), and AWS, ensuring high availability, system security, and low-latency API communication.
- **System Automation & API Integration:** Built automated scripts in C# and VB.NET to reduce manual business processes and connected Epicor ERP with external third-party logistics and manufacturing monitoring systems via RESTful middleware.
- **Key Project:** Accounts Receivable Portal, IPAK5 (PWA, IOS App, Android App), Nexfusion (PWA), TGxNinja (PWA), ITRequisitionForm, Firm Performance Test (PWA), Sales Report Dashboard, Auto Signature and Stamp Cop in SRSS Report, Courier Service Requisition System (PWA), Production Dashboard System,

Test Engineer 1 — Jul 2022 - April 2024

Jabil Circuit Sdn Bhd (P1) | Bayan Lepas, Penang, Malaysia

- **Data Integrity & Dashboard Validation:** Developed and executed comprehensive test plans to validate SMT production data pipelines, machine outputs, and data transformations feeding into executive Power BI dashboards.
- **Automated Data Quality Testing:** Designed and maintained automated validation scripts to verify reporting accuracy, data consistency, and manufacturing KPI metrics across underlying datasets.
- **Embedded IoT System Development:** Built and deployed a custom Raspberry Pi power-monitoring system, integrating hardware sensors and data acquisition modules to capture real-time electricity tariff consumption during chamber test cycles.
- **Performance Optimization:** Conducted performance and latency analysis on Power BI dashboards and SMT data pipelines, identifying data bottlenecks to optimize reporting efficiency.
- **Cross-Functional Collaboration:** Partnered with manufacturing engineers, BI developers, and quality teams to ensure raw production metrics aligned with core business and reporting requirements.
- **Test Documentation & Quality Control:** Maintained structured test cases, automated verification logs, and defect reports for SMT data systems, supporting continuous improvement in data reliability.
- **Key Project:** Chamber Utilization System and Connect with Raspberry PI

Jabil Circuit Sdn Bhd (P2) | Bayan Lepas, Penang, Malaysia

- **Test Script Development:** Programmed and executed functional test scripts using Python in Linux environments to validate 3 New Product Introduction (NPI) projects for Cisco Systems.
- **Automation & Execution:** Developed test code and executed validation sequences using PyCharm and Apollo GUI, ensuring test coverage aligned with customer product specifications.
- **Failure Analysis & Debugging:** Conducted root cause analysis on failing units, collaborating with design teams and senior test engineers to debug code, fix test logic errors, and resolve hardware/software integration issues.
- **Version Control & CI/CD Pipelines:** Managed test code repositories in Bitbucket and integrated automated script execution via Jenkins.

- **Hardware & Fixture Support:** Assisted lead engineers with test fixture debugging, validation, and setup, ensuring test routines ran reliably on prototype units.
- **Documentation & Build Traceability:** Authored test execution notes and tracked bill-of-materials (BOM) configurations using MobaXterm, Cesium, and enterprise BOM systems.
- **Key Project:** NPI Project (Vigor LS, Vigor LS SE, La Corsa)

Software Engineer — June 2021 - May 2022

Hospital Canselor Tuanku Muhriz UKM - HCTM | Cheras

- **Mobile Full-Stack Development:** Engineered a native Android mobile application using Java, Android Studio, and RESTful APIs, enabling emergency ward clinicians to capture real-time patient status and prioritize surgical cases.
- **Computer Vision & Hardware Integration:** Implemented camera-based Medical Record Number (MRN) scanning features to dynamically query and populate patient profiles, eliminating manual data entry errors during critical triage.
- C**linical Workflow Automation:** Translated emergency clinical triage criteria and vital sign parameters into automated decision-support logic for priority slotting in operating theatres.
- **Backend Architecture & Database Optimization:** Built backend services using .NET Framework (ASP.NET) and MySQL; utilized SQL Server Profiler to analyze execution queries and optimize database response times under high-frequency ward usage.
- **System Interoperability & Platform Integration:** Integrated the OTIS web platform with the ESB mobile application via REST endpoints, delivering real-time visibility of emergency surgical requests to all operating room staff and surgical teams.
- **Resource Scheduling & Conflict Detection:** Analyzed OT workflow requirements to build dynamic scheduling modules featuring automated slot conflict detection, instrument availability tracking, and automated surgical list generation.
- **Real-Time Telemetry & Compliance:** Programmed automated timestamping modules (patient admission, anesthesia, surgery start/end, recovery) to establish end-to-end audit trails for compliance and operational bottleneck analysis.
- **Key Project:** Emergency Surgery Booking System (ESB), Operation Theathre Information System (OTIS)

PROJECT

iPAK5 Platform (PWA, iOS, Android, Huawei)

- **Cross-Platform Mobile Ecosystem:** Developed a responsive PWA and native mobile applications (iOS/Android/Huawei) utilizing C#, ASP.NET, Java, Swift, Python, and Tailwind CSS.
- **IoT & Camera Integration:** Integrated QR scanning for instant film specification lookup and built edge telemetry features with Raspberry Pi for real-time machine status monitoring.
- **AI & Cloud Infrastructure:** Built an integrated AI chat assistant for machine diagnostics and deployed data pipelines across AWS (EC2, S3, RDS) with offline-first synchronization.

NexFusion Customer Portal (PWA)

- **ERP & Machine Data Aggregation:** Architected a unified PWA linking Epicor/Kinetic ERP APIs and ESTL machine databases into a single customer-facing portal.
- **Self-Service Analytics:** Enabled client visibility into real-time product specs, order fulfillment, job telemetry, firm test results, and automated invoice downloads.
- **Modern Stack:** Built backend REST services using C# and ASP.NET with a responsive Tailwind CSS and JavaScript frontend.

PROJECT

TGxNinja (PWA) - TGPT

- Automates courier form submission by integrating NinjaVan, SiteGiant, and Kinetic APIs.
- Auto-retrieves customer and shipment details, eliminating manual data entry.
- Reduces processing time and minimizes errors.
- Built with ***Asp.Net, C#, Javascript,Tailwind CSS, Node.js. Visual Studio, REST API.***

Firm Performance Test (PWA) - TGPT

- Retrieves machine data and firm test results by scanning QR codes.
- Integrated with ESTL API and Kinetic API.
- Displays graphs, trends, and historical performance data.
- Enables quick mobile access on the production floor.
- Built with ***Asp.Net, C#, Javascript,Tailwind CSS, Node.js. Visual Studio, REST API.***

Courier Service Requisition System (PWA) - TGPT

- Digitized courier request and approval process.
- Real-time shipment tracking via courier API integration.
- Role-based access and automated notifications.
- Built with ***Asp.Net, C#, Javascript,Tailwind CSS, Node.js. Visual Studio, REST API.***

Automated Accounts Receivable & Payment Gateway Portal (PWA)

- **Financial API Integration:** Integrated Maybank Payment Gateway APIs with Kinetic ERP OData REST APIs using ASP.NET in PWA mode to automate customer payment reconciliation.
- **Automated ERP Cash Entry:** Engineered a single-click workflow that verifies customer bank transfers in real time and automatically posts matched Cash Receipt Entries directly into Kinetic ERP, eliminating manual financial data entry and reducing payment posting cycles by 90%.

Migration Epicor to Kinetic (ERP System) - TGPT

- Assisted in migrating ERP from Epicor to Kinetic.
- Managed data transfer, SSRS report path updates, and configuration changes.
- Performed validation and testing for seamless continuity.
- Received Appreciation Award for successful migration.
- Convert SSRS Report to Live.

Chamber Utilization System and Connect with Raspberry PI - Jabil P2

- Raspberry Pi monitors electricity tariffs during peak/non-peak hours.
- Data collected via Modbus/RS485 and stored in MySQL.
- Power BI dashboards visualize usage and energy costs.
- Helped reduce energy costs by 15–20%.
- Built with **C#, SQL Server, VB.Net, Visual Studio, RaspberryPi, Linux, Pyhton.**

NPI Project - Jabil P1

- Led NPI testing for 3 Cisco projects: Vigor LS, Vigor LS SE, La Corsa.
- Defined test strategies, performed hardware validation, and managed risks.
- Maintained >95% production yield.
- Ensured mass production readiness.
- Built with ***Python, Linux Shell Scripting, Bitbucket, PyCharm, Jenkins, Apollo GUI, BOM, MobaXTerm.***

EDUCATION

UNITAR International University (UNITAR) **Oct 2026 - Present**

- Master of Information Technology

The National University of Malaysia (UKM) **Sept 2018 - Sept 2022**

- Bachelor of Software Engineering Major in Information System Development
- **CGPA: 3.62**
- Dean List: Sem 2 Session 2019/2020, Sem 1 Session 2020/20221, Sem 2 Session 2020/2021, Sem 2 Session 2021/2022

CERTIFICATIONS & PROFESSIONAL AFFILIATIONS

- Graduate Technologist (Gt.) – Information & Computing Technology (IT) | Malaysia Board of Technologists (MBOT) | ***Cert No: GT26050659***
- Certified Professional Technologist (Ts.) in Information & Computing Technology (IT) | Malaysia Board of Technologists (MBOT) | ***PT26100356***
- IBM Full Stack Software Developer Professional Certificate | ***Cert No:JZQV14M3ZNVO***
- IBM Systems Analyst Professional Certificate | ***Cert No: M9TT2IPMHM2E***
- Microsoft SQL Server Professional Certificate | ***Cert No: F8NKHDSDRA4J***
- Cisco Implementation and Administration (CCNA 200-301) | ***Cert No: 0CUYL4R4Z2C2***
- Linux Administration and Shell Scripting Specialization Certificate | ***Cert No: MDE0KL2MJ291***
- Intermediate Oracle Database Administration | ***Cert No: VZRJA46PSFWD***
- Introduction to Business Analysis | ***Cert No: 2GYBXH0JLOC6***
- JBL-LEAN-Bronze Six Sigma | ***Cert No: 8146a1e5-1edc-44a6***
- EMS-Malaysia-Penang-QE-CAPA System (SmartSolve) Rev E

HONORS & AWARDS

- **Appreciation Award (2025)** – Thong Guan Technologies Sdn. Bhd. (TGPT) | Awarded for successfully supporting and executing the enterprise-wide migration from Epicor ERP to Kinetic with zero operational downtime.
- **Best FYP Award (Supervisor Recognition)** – Universiti Kebangsaan Malaysia (UKM) | Recognized for outstanding software engineering design and research on the Celik Pusaka mobile application.
- **Dean's List Award (4 Consecutive Semesters)** – FTSM, Universiti Kebangsaan Malaysia (UKM) | Achieved academic excellence across four semesters maintaining a high CGPA.

LANGUAGE

- **English** : Proficient **Malay** : Proficient

REFERENCES

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